

Hands-On sessions: Linking motifs to variant effects



ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

Kilian Salomon
Computational Genome Biology @ BIH

July 20, 2025



UNIVERSITÄT ZU LÜBECK



UNIVERSITÄTSKLINIKUM
Schleswig-Holstein



Hands-on: Variant Effects and Motifs



https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial/tree/main/06_variant_effects_and_motifs

1. Introduction: Variants Tested with MPRA

- **Data sources:** MPRAsnakeflow results, Variant map, BCalm variant effects
- **Steps:**
 - i. Normalize counts
 - ii. Run BCalm using normalized counts and variant map
 - iii. Identify significant variants

2. Transcription Factor Binding Motif Identification

- **Databases:**
 - JASPAR2022
 - HOCOMOCO v13
- **Tools:**
 - HOMER
 - FIMO (MEME suite)
- **Workflow:**
 - Run FIMO on test data
 - Merge TFBS information with variant metadata
 - Assess if TFBS are gained or lost due to variants
 - Increase confidence in TFBS calls (e.g., by filtering for cell-type expression)
 - Filter for variant-overlapping TFBS
 - Visualize the proportion of variants overlapping TFBS

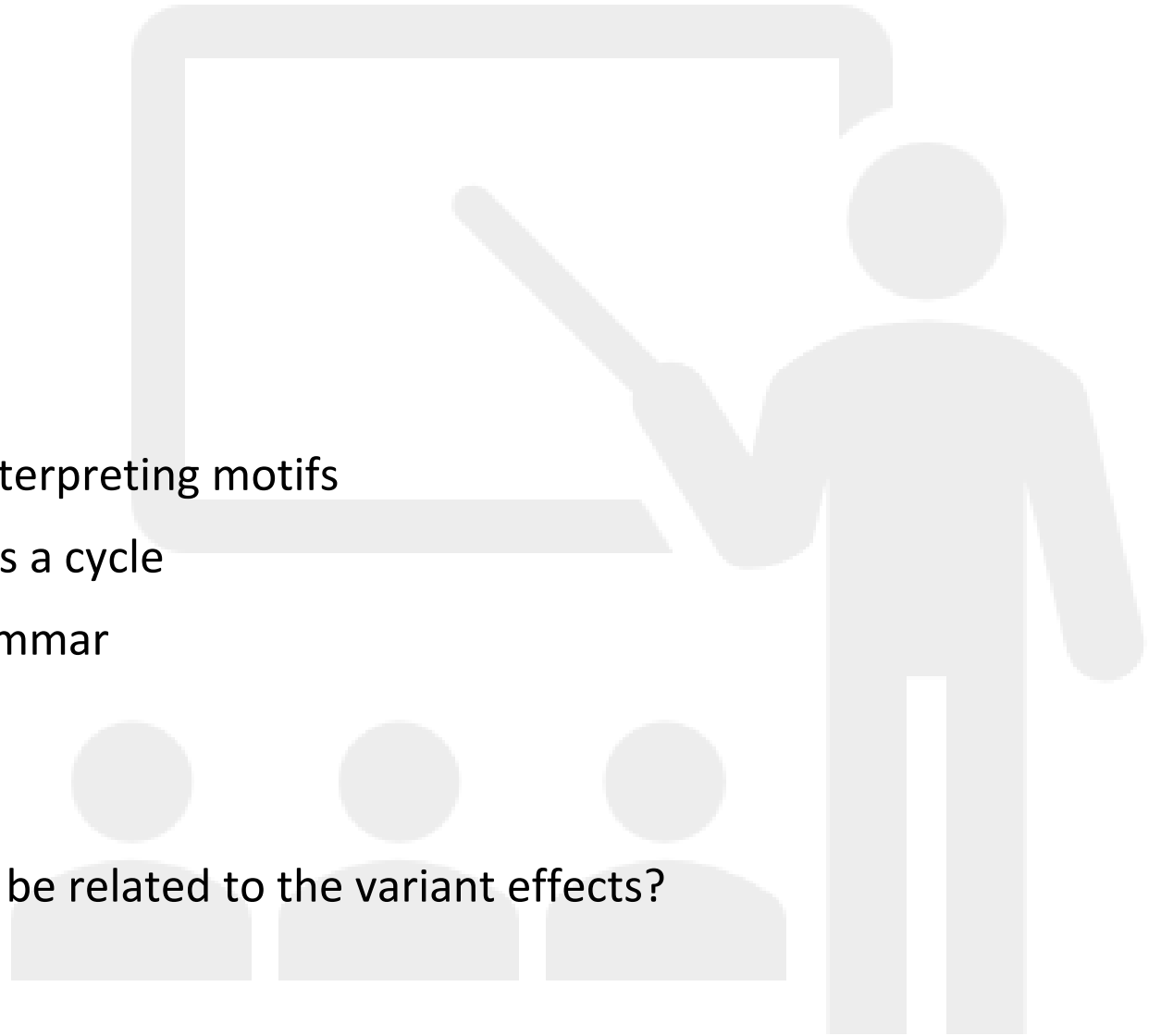
Scope of today's workshop

Goals

- Investigating variant-based dataset
- Get to know a motif database
- Learn about workflow of finding and interpreting motifs
- Functional validation in living systems is a cycle
- Glimpse on the complexity of TFBS grammar

Questions we will cover:

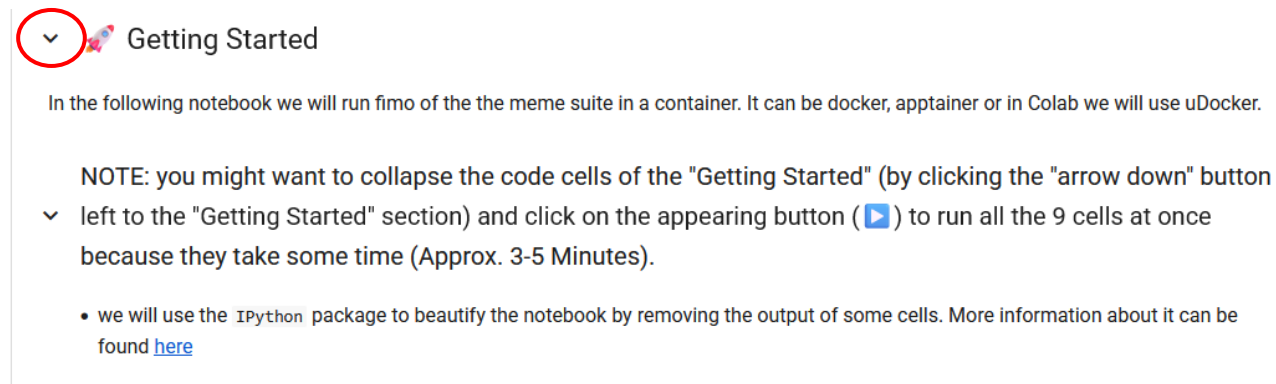
- Which known transcription factors can be related to the variant effects?



Getting started

The installation of the required packages takes some time so let's start it first

- Please collapse the „Getting Started“ section and run all cells at once
 - By clicking on the marked arrow
 - Followed by clicking the run button

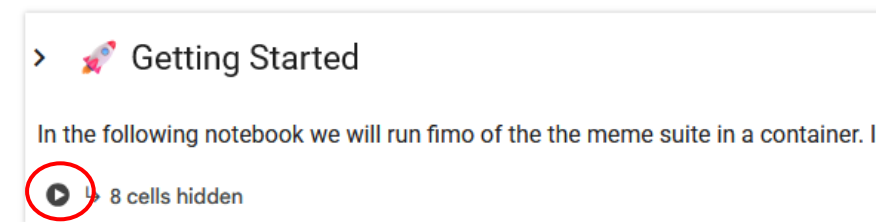


Getting Started

In the following notebook we will run fimo of the the meme suite in a container. It can be docker, apptainer or in Colab we will use uDocker.

NOTE: you might want to collapse the code cells of the "Getting Started" (by clicking the "arrow down" button left to the "Getting Started" section) and click on the appearing button (▶) to run all the 9 cells at once because they take some time (Approx. 3-5 Minutes).

- we will use the `IPython` package to beautify the notebook by removing the output of some cells. More information about it can be found [here](#)



> Getting Started

In the following notebook we will run fimo of the the meme suite in a container. I

▶ 8 cells hidden

Short intro into the variant dataset used



Theofilos
Chalkiadakis

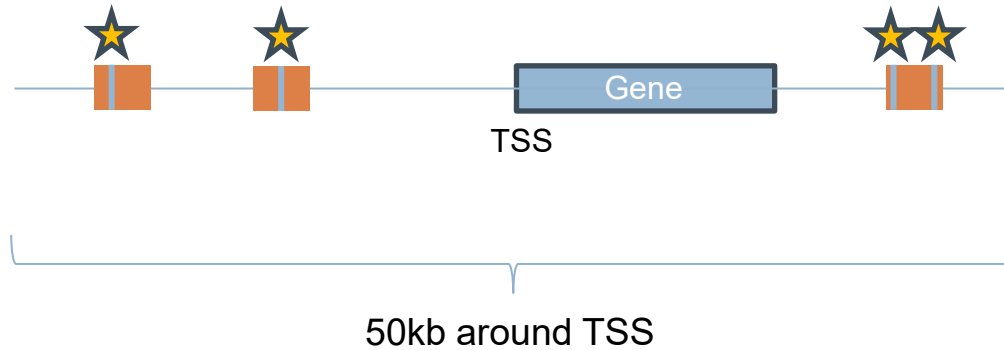


Mohan
Dash



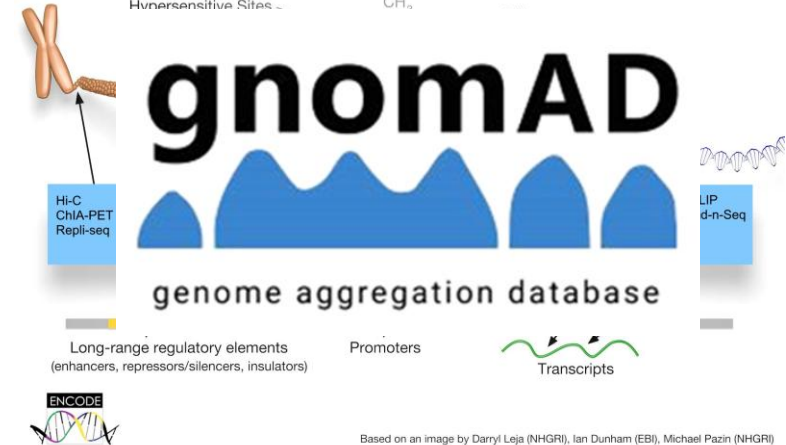
Max
Schubach

Define candidate regulatory sequences



SCREEN: Search Candidate cis-Regulatory Elements by ENCODE

Registry of cCREs V3

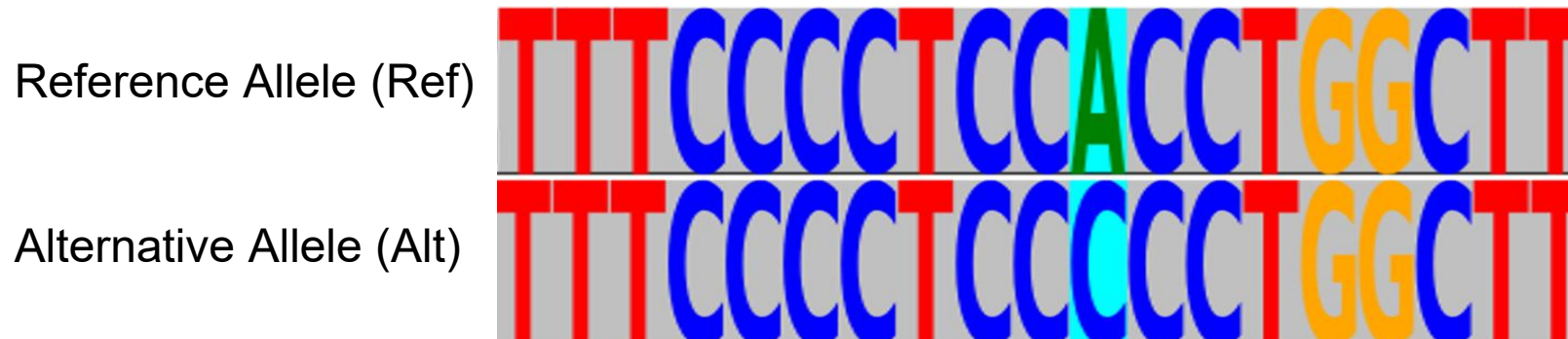


Known variation in cis-regulatory regions of a 50 kb window around disease associated genes

Motif Identification from Sequence

GOAL: Identify potential TFBS which binding affinity might be affected by the variant of interest

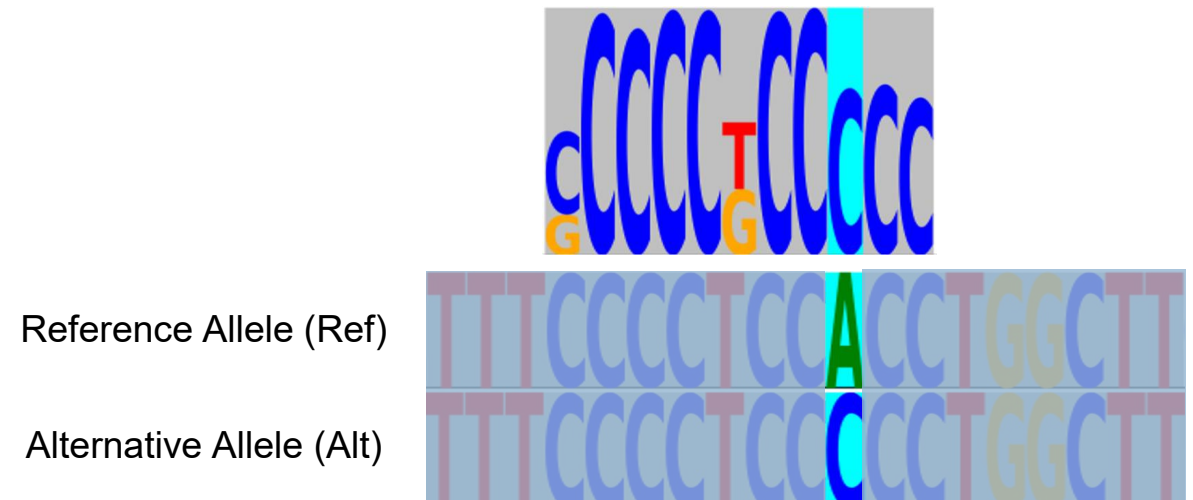
- We have reference and alternative allele
- We are interested in TFBSs



Motif Identification from Sequence

GOAL: Identify potential TFBS where binding affinity might be affected by the variant of interest

- We have reference and alternative allele
- We are interested in TFBSs
- The height of the character represents the „importance“ for the TFBS
- Let's focus on the variant



Where does the TFBS fit best?

Is the motif fitting more likely the reference or alternative allele?

First: The C of the alternative allele increases the fit of the motif

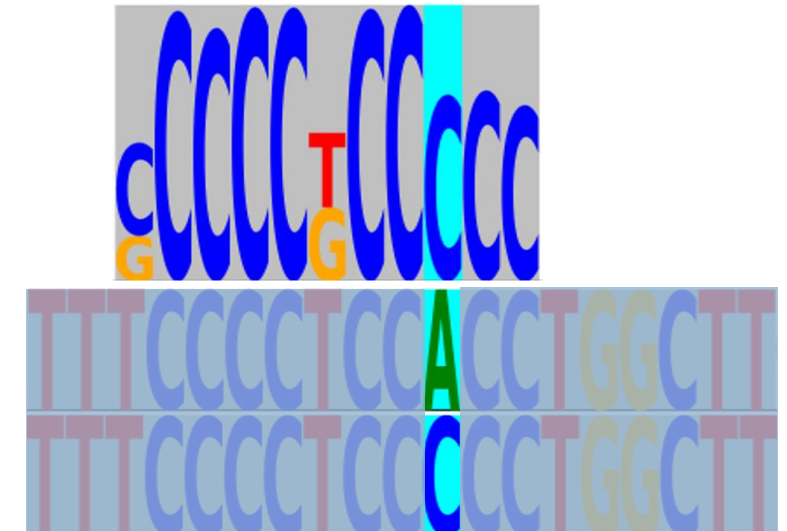
- This motif is predicted to be gained because of the variant

Second: The C of the reference allele increases the fitting of the motif

- This motif is predicted to be lost because of the variant

What do we need?

- Set of potential TFBS we are interested in



TFBS motif databases

Multiple options exist and depend on your task/data

Database	Focus / Scope	Source Organisms	Format	Notes
JASPAR	Non-redundant curated TF motifs	Primarily vertebrates, plants, insects	MEME, JASPAR	Open-access; manually curated; widely used in academia
HOCOMOCO	High-quality TF motifs from ChIP-seq	Human, mouse	MEME, TRANSFAC	High-quality, ChIP-seq-based motifs; ranking by quality
TRANSFAC	Comprehensive TF–DNA interactions	Multiple (mainly human)	Proprietary	Commercial license required; extensive annotations
CIS-BP	Predicted and experimental motifs	600+ species	MEME, custom	Combines multiple sources; includes inferred motifs
Factorbook	ENCODE-derived TF motifs	Human	JSON, BED	Motifs directly from ENCODE ChIP-seq data; ENCODE integration

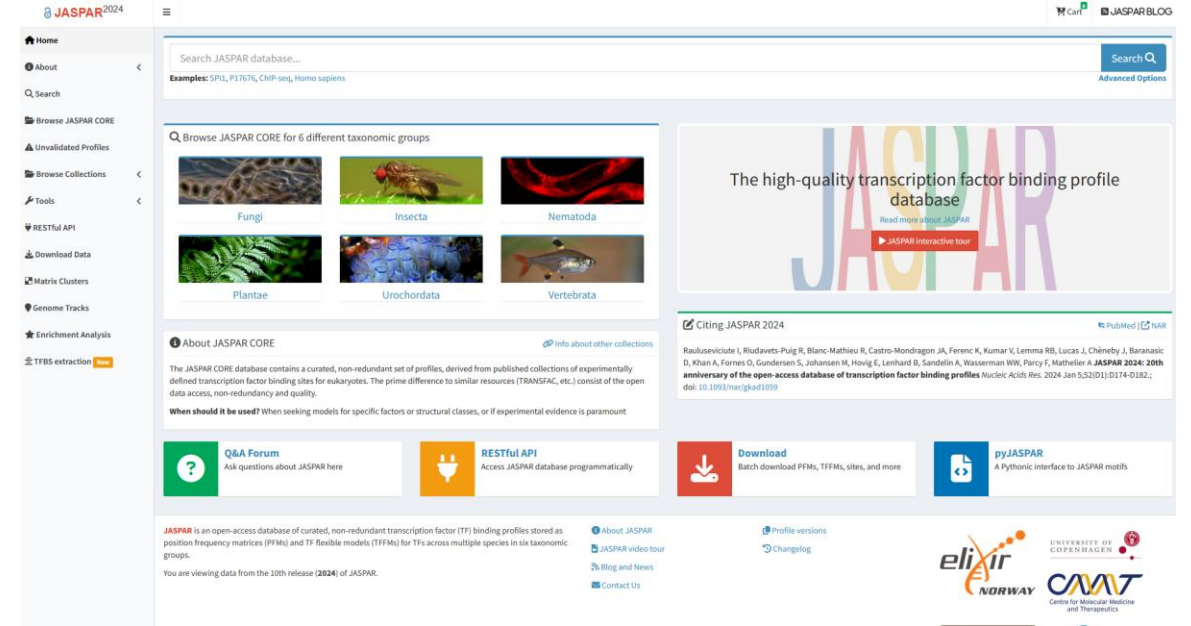
TFBS motif databases: widely used in academia

JASPAR

- Open access and manually curated motifs
- Broad cross-species coverage
- Compatibility with most tools (e.g., FIMO, HOMER)
- In 2024 CORE: 2346

HOCOMOCO

- Open access and manually curated motifs
- Human/mouse-focused, quality-ranked motifs
- Compatibility with most tools (e.g., FIMO, HOMER)
- In v13 CORE: 1611



HOCOMOCO v13									
v13 Core collection (complete)									
Motif	LOGO	Gene (human)	Gene (mouse)	Quality	Motif subtype	Data sources	TF family	UniProt ID (human)	UniProt ID (mouse)
AHCTF1.H13CORE.0.B.B	AAAT	AHCTF1 (GeneCards)	Ahctf1	B	0	B	Unannotated [8.2.255] (TFClass)	ELYS_HUMAN	ELYS_MOUSE
AHR.H13CORE.0.B	TCGTG	AHR (GeneCards)	Ahr	B	0	P	PAS [1.2.5] (TFClass)	AHR_HUMAN	AHR_MOUSE
AHRR.H13CORE.0.RC	TCGTG	AHRR (GeneCards)	Ahrr	C	0	P	PAS [1.2.5] (TFClass)	AHRR_HUMAN	AHRR_MOUSE
ALX1.H13CORE.0.SM.B	TAATTTA	ALX1 (GeneCards)	Alx1	B	0	S-M	Paired-related HD [3.1.3] (TFClass)	ALX1_HUMAN	ALX1_MOUSE
ALX3.H13CORE.0.SM.B	TAATT	ALX3 (GeneCards)	Alx3	B	0	S-M	Paired-related HD [3.1.3] (TFClass)	ALX3_HUMAN	ALX3_MOUSE
ALX3.H13CORE.1.S.B	TAATTATTA	ALX3 (GeneCards)	Alx3	B	1	S	Paired-related HD [3.1.3] (TFClass)	ALX3_HUMAN	ALX3_MOUSE
ALX4.H13CORE.0.S.B	TAATTA	ALX4 (GeneCards)	Alx4	B	0	S	Paired-related HD [3.1.3] (TFClass)	ALX4_HUMAN	ALX4_MOUSE
ALX4.H13CORE.1.SM.B	TAATT	ALX4 (GeneCards)	Alx4	B	1	S-M	Paired-related HD [3.1.3] (TFClass)	ALX4_HUMAN	ALX4_MOUSE

„Compatible with most tools“

Identifying given TFBS in a sequence set can be done using different options, again.

FIMO (Finding Individual Motif Occurrences)

- Scan sequences for motif matches
- Returns statistically significant motif hits (p/q-values)

HOMER (Hypergeometric Optimization of Motif EnRichment)

- Scan & discover (new) motifs
- Enrichment between two sets of sequences
- Less transparent statistical information about the resulting hits

We will just use one tool: FIMO

Input:

- Sequences as fasta and the motif database (HOCOMOCO v13)

Output:

- Table with motif id with a p-value and FDR corrected q-value (higher confidence = lower)

Column	Description
motif_id	ID of the matched motif
sequence_name	Name of the sequence with the motif
start/stop	Start/stop positions of the motif match
strand	Strand (+/-)
matched_sequence	Sequence matching the motif
p-value	Motif match p-value (default threshold: 1e-4)
q-value	FDR-corrected p-value (higher confidence = lower)

Now keep exploring the dataset

Now you will...

- ... explore the data
- ... identify the TFBS which overlap the variants
- ... identify how many overlapping TFBS we find
- ... check how well your eyes are at this task

Please mention when you are at the task to compare the TFBS by eye

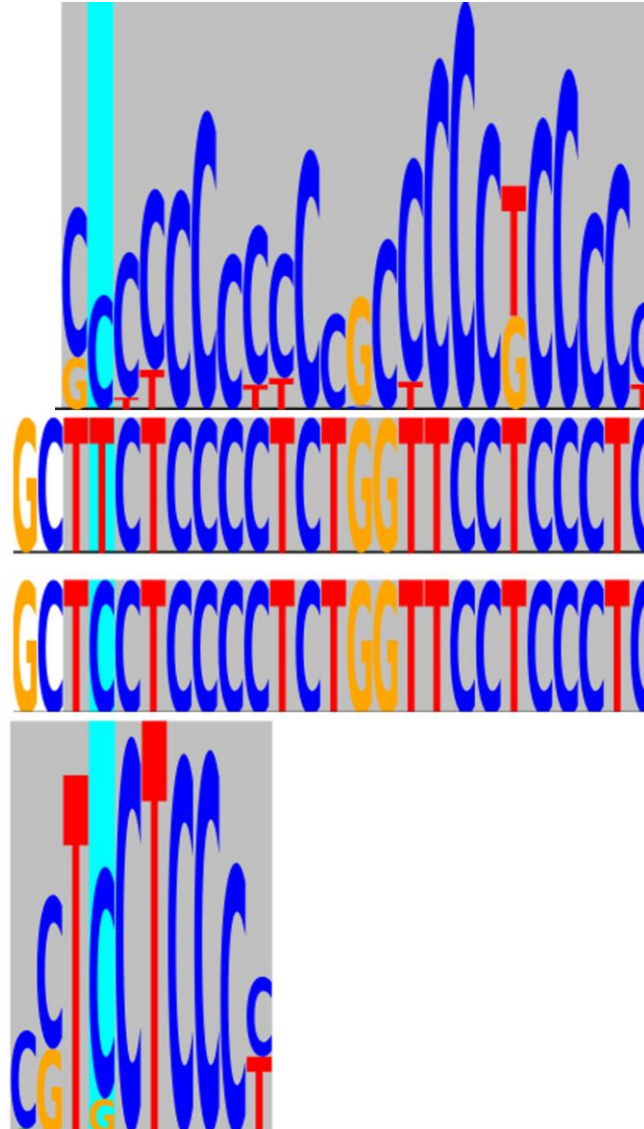
Which TFBS fits better? Eye vs q-value

PATZ1.H13CORE.0.P.B (rev)
Fimo q-value (Alt): 0.006

Ref

Alt

ZN263.H13CORE.0.PM.A (rev)
Fimo q-value (Alt): 0.0602



Now you will...

- ... classify the TFBS based on which allele they fit better
- ... investigate the variant effect of the tested variants
- ... interpret if the identified Motif is predicted to activate or repress

On which allele does the predicted TFBS fit more likely?

A TFBS can be found either on both reference and alternative allele or only in one of them, depending on the result from FIMO

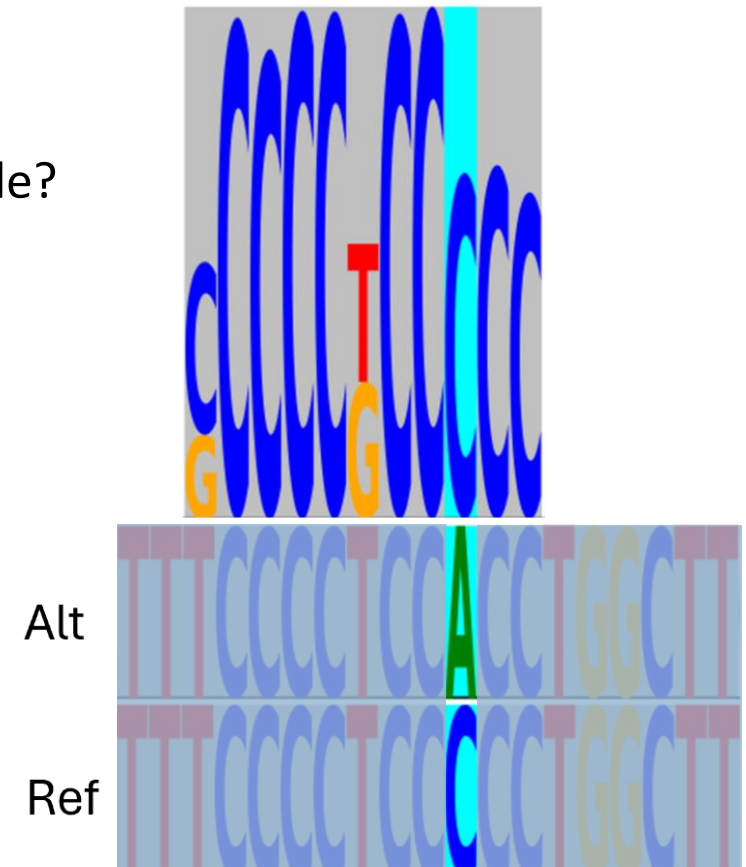
The question can be rephrased to:

- Is the motif fitting more likely the reference or alternative allele?

How many TFBS are predicted to fit more likely

- ... on the alternative compared to the reference allele (gain)?
- ... on the reference compared to the alternative allele (loss)?

TFBS change	Number of motifs	Number of variants
gain	12	8
loss	10	8
No change	4	3



Interpret variant effect on TFBS level

Activating TFBS:

- if ALT binds better and positive effect => motif is predicted to increase transcription in this cell type
- if REF binds better and negative effect => motif is predicted to increase transcription in this cell type

Repressing TFBS:

- if ALT binds better and negative effect => motif is predicted to decrease transcription in this cell type
- if REF binds better and positive effect => motif is predicted to decrease transcription in this cell type

TFBS interpretation	count
Predicted transcriptional increasing TFBS	12
Predicted transcriptional decreasing TFBS	10

You might ask: is a gain of a TFBS always transcription increasing?

TFBS change	Number of motifs
gain	12
loss	10
No change	4

TFBS interpretation	count
Predicted transcriptional increasing TFBS	12
Predicted transcriptional decreasing TFBS	10

If we group by the TFBS interpretation, we can see that this is not true

TFBS interpretation	Gain	Loss
Predicted transcriptional decreasing TFBS	4	6
Predicted transcriptional increasing TFBS	8	4

Conclusion

Do not trust the results of TFBS predictions without sanity checking

Being more stringent might reduce the number of false positives

So does using additional information

- Focusing on a reasonable subset of TFs
- Focusing on only the well-fitting motifs

Obviously, a gained motif does not need to increase the transcriptional activity

Interpretations of these motifs do not proof actual binding

- Functional validation is the next step

Hocomoco naming convention

ALX4.H13CORE.1.SM.B and ALX4.H13CORE.0.S.B

- ALX4: Uniprot accession
- H13CORE: Motif set
- .1 or .0: subtype number
 - .0 primary motif
 - .1 (or .2) alternative binding motifs for the same TF (with experimental support)
- .SM or .S: experimental data type
 - P: ChIP-Seq (Chromatin Immunoprecipitation Sequencing)
 - S: HT-SELEX (High-Throughput Systematic Evolution of Ligands by Exponential Enrichment)
 - M: Methyl-HT-SELEX (HT-SELEX with methylated CpGs)
- .B: motif quality
 - A-quality: found by both HT-SELEX and ChIP-Seq
 - B-quality: found in at least 2 different experiments; C-quality: single experiment + expert curation
 - D-quality: not tested, usually inherited from earlier version, Not re-discovered, kept for consistency



Discussing data limitations

ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

Arjun Devadas, Kilian Salomon, Martin Kircher, Max Schubach
Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

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of Health
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This session is an opportunity to discuss the boundaries of current MPRA technology and modeling approaches and identify areas where more research or methodological development is needed.

- Share Your Challenges: Can you share specific examples from your own research where you encountered limitations related to variant effects, sequence context, or cell-type specificity?
- Interpretation of MPRA Data: What do MPRA limitations mean for drawing robust conclusions from MPRA experiments?
- Implications for Predictive Modeling: How do data limitations impact the accuracy and generalizability of models trained on MPRA data, and how these models can be improved to account for cell-type specificity or complex sequence contexts?
- Importance of the genomic context: MPRA activities are often tested outside their native genomic context and are difficult to translate into phenotypic or organismal effects. What are strategies to better connect in-vitro MPRA findings to a broader biological and clinical relevance?
- Design of future MPRA: How do current limitations inform the design of future MPRA experiments or computational approaches to overcome current hurdles, such as expanding cell-type diversity or improving sensitivity to subtle cell-type specific effects?

Thank You! Questions?

kircherlab.github.io

CONTACT

Regulatory Genomics & Computational Genome Biology

Martin Kircher martin.kircher

[@uni-luebeck.de](mailto:martin.kircher@uni-luebeck.de) | [@uksh.de](mailto:martin.kircher@uksh.de) | [@bih-charite.de](mailto:martin.kircher@bih-charite.de)



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Participant questions and feedback

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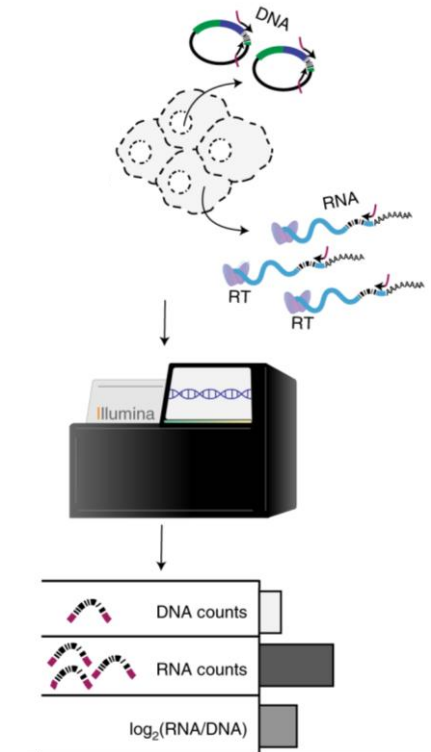
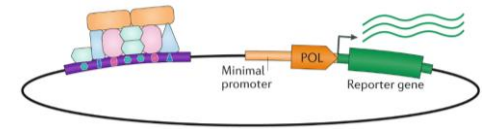


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Revisiting our Learning Objectives

- Introduction to IGVF and how to access its data.
- MPRA Experiments: Explanation of experimental design, including the association of barcode/tag sequences to tested regulatory sequences and variants.
- Step-by-step data processing using MPRAsnakeflow, including construction of count tables and interpretation of quality control metrics.
- Statistical analysis of DNA and RNA barcode counts using BCalm.
- Applications of MPRA data like identifying active regulatory regions and significant variant effects, optimising cell-type-specific expression through synthetic sequences.
- Sequence-based modeling of regulatory sequence activity, i.e. using deep learning models to predict open chromatin and MPRA activity, investigate transcription factor motif importance and composition.



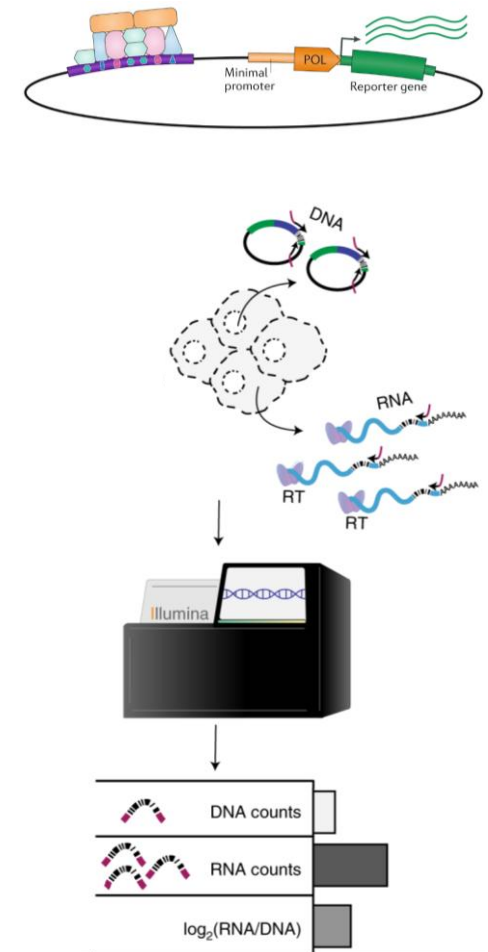
Questions?



Questions for you? (1)

Reflecting on your motivation for attending this tutorial, **did the content covered** – from the introduction to IGVF and MPRA experiments to data processing, analysis, and applications – **meet your expectations?**

Which specific learning objective, such as understanding MPRA experiments, mastering data processing steps with MPRAsnakeflow, or exploring applications like variant identification or sequence modeling, **did you find most valuable for your current or future research?**

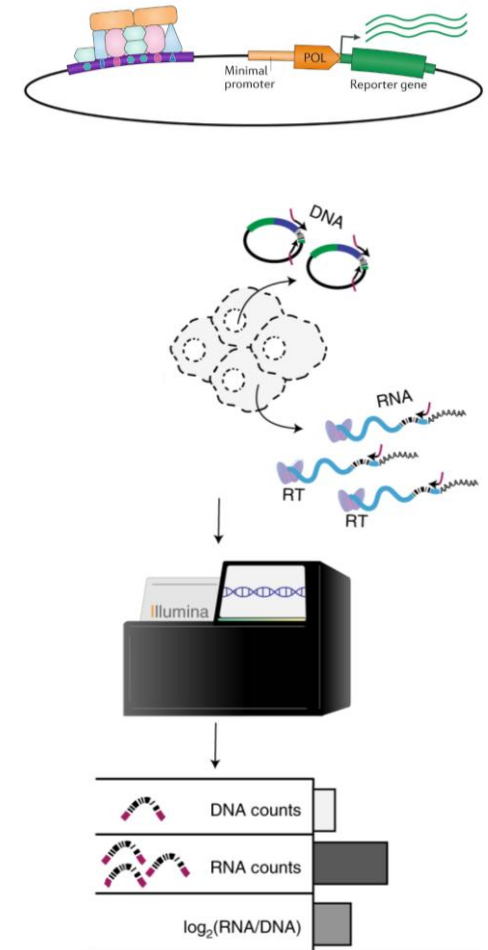


Questions for you? (2)

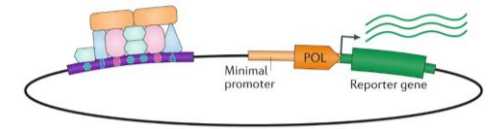
What part of the tutorial content or hands-on practice do you believe will have the most immediate impact on your research in functional genomics, variant analysis, or sequence model training?

How confident do you feel about starting to process and analyze MPRA data using tools like MPRAsnakeflow or BCalm after participating in the hands-on exercises on Google Colab?

What further support might help increase your confidence?

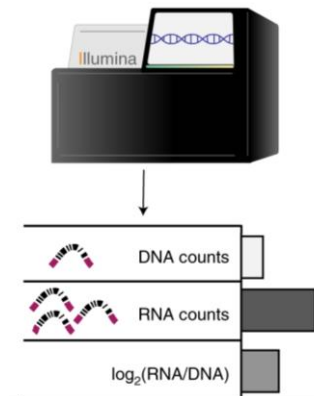
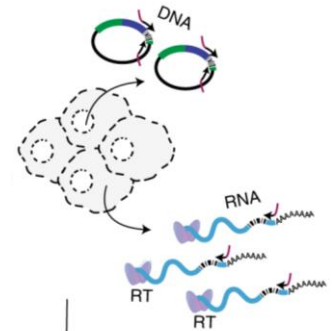


Questions for you? (3)



How useful was the introduction to the iGVF consortium and learning how to access its data resources?

Do you see opportunities to leverage iGVF data in your work?



Thanks to everyone to contributing and making this a successful day!

- **Thank you to all participants for their active role!**
- **Thank you to our speakers:**
 - Arjun Devadas, University of Lübeck / UKSH
 - Kilian Salomon, Berlin Institute of Health at Charité
 - Martin Kircher, University of Lübeck / UKSH & Berlin Institute of Health at Charité
 - Max Schubach, Berlin Institute of Health at Charité
- **And everyone supporting the tutorial preparation:**
 - Michael Love, University of North Carolina
 - Jonathan Rosen, University of North Carolina
 - Pia Keukeleire, University of Lübeck / UKSH
 - Pyaree-Mohan Dash, Berlin Institute of Health at Charité
 - Thorben Maaß, University of Lübeck / UKSH
 - **ISMB-ECCB reviewers and Tutorial Organizers**



Please participate in the ISMB-ECCB tutorial feedback!

Share your Feedback!

**Scan the QR code to let us
know your thoughts on
this tutorial. Thank you!**



Feel free to reach out!

kircherlab.github.io

CONTACT

Regulatory Genomics & Computational Genome Biology

Martin Kircher martin.kircher

[@uni-luebeck.de](mailto:martin.kircher@uni-luebeck.de) | [@uksh.de](mailto:martin.kircher@uksh.de) | [@bih-charite.de](mailto:martin.kircher@bih-charite.de)



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